Exercise for Chapter 15

Ke Hua (花 珂)*

State Key Laboratory of Solidification Processing,
Northwestern Polytechnical University,
Xi'an, Shaanxi, P.R. China

*ke.hua@nwpu.edu.cn

http://teacher.nwpu.edu.cn/kehua

15 .1* 2-4 Landau Case:

The excess Gibbs free energy as a function of order parameter for a solution is written as

$$G^{XS} = G_{\text{ord}} - G_{\text{dis}} = a\left(T - T_C\right)\eta^2 + C\eta^4$$

where G_{dis} is the free energy of the disordered phase and *a* and *C* are positive constants.

a. Obtain an expression for the excess entropy of the equilibrium ordered phase as a function of temperature.

b. Determine the value of $\Delta C_P = C_P{}^{\text{ord}} - C_P{}^{\text{dis}}$ at the transition temperature T_C.

Solution to 15.1

(a)

$$G^{\mathrm{XS}} = G_{\mathrm{ord}} - G_{\mathrm{dis}} = a\left(T - T_C\right)\eta^2 + C\eta^4$$

First we obtain $\eta_{\mathrm{eq}}^2 = -\dfrac{a\left(T - T_C\right)}{2C}$ by setting $\dfrac{\partial G^{XS}}{\partial \eta} = 0$ and solving for η_{eq}.

$$G^{\mathrm{XS}} = a\left(T - T_C\right)\eta_{\mathrm{eq}}^2 + C\eta_{\mathrm{eq}}^4 = -\frac{a^2\left(T - T_C\right)^2}{4C}$$

$$\frac{\partial G^{\mathrm{XS}}}{\partial T} = -S^{XS} = -\frac{2a^2\left(T - T_C\right)}{4C}$$

$$S^{XS} = \frac{2a^2\left(T - T_C\right)}{4C} \quad \text{(valid only at } T < T_C\text{)}$$

S^{XS} is negative since it is the excess entropy over and above the disordered entropy.

Solution to 15.1

(b)

$$C_P = T\left(\frac{\partial S}{\partial T}\right)_P$$

Thus ,since

$$S_{\mathrm{ord}} - S_{\mathrm{dis}} = \frac{a^2\left(T - T_C\right)}{2C}$$

$$\frac{\partial S_{\mathrm{ord}}}{\partial T} - \frac{\partial S_{\mathrm{dis}}}{\partial T} = \frac{a^2}{2C} \qquad T_C\frac{\partial S_{\mathrm{ord}}}{\partial T} - T_C\frac{\partial S_{\mathrm{dis}}}{\partial T} = \frac{T_C a^2}{2C} \qquad C_P^{\ \mathrm{ord}} - C_P^{\ \mathrm{dis}} = \frac{T_C a^2}{2C} > 0$$

$$\text{and since} \quad T_C = \frac{2C}{a}, \quad C_P^{\ \mathrm{ord}} - C_P^{\ \mathrm{dis}} = a$$

The heat capacity is of the ordered phase is larger than that of the disordered phase since some thermal energy must be used to disorder the ordered phase on heating. See Figure 15.15.

15.2 * 2-4-6 Landau Case

The Gibbs energy as a function of order parameter for a solution is written as

$$G = a\left(T - T_C\right)\eta^2 + C\eta^4 + E\eta^6$$

For this case, assume $C < 0$ and a and E are positive.

a. Find the nonzero value of the order parameter of the solution that has the same Gibbs energy as that of the disordered solution.

b. Sketch the Gibbs energy versus η curve for the temperature in question in (a). This temperature can be called T_0.

c. Determine if the transformation for this alloy is first order. Explain.

d. Calculate the heat of transformation, ΔH_0 , for this disorder/order transformation in terms of (a), η_0 and T_{tr}, where η_0 is the order parameter at the equilibrium transition temperature T_0.

e. What is the significance of the sign of ΔH_{tr} for this transformation?

Solution to 15.2

(a)

$G^{\mathrm{XS}}_{\mathrm{disorderd}}=0$ Thus *G* of the ordered phase also = 0.

If equilibrium obtains: $\dfrac{\partial G}{\partial \eta}=0$

Thus we have 2 equations to solve:

$$G = a\left(T - T_C\right)\eta^2 + C\eta^4 + E\eta^6 = 0$$

$$\frac{\partial G}{\partial \eta} = 0 = 2a\left(T - T_C\right)\eta + 4C\eta^3 + 6E\eta^5 = a\left(T - T_C\right)\eta^2 + 2C\eta^4 + 3E\eta^6$$

Solving these two equations:

on eliminating E we obtain: $\eta = 0 \text{ and } \eta^2 = -\dfrac{2a(T - T_C)}{C}$

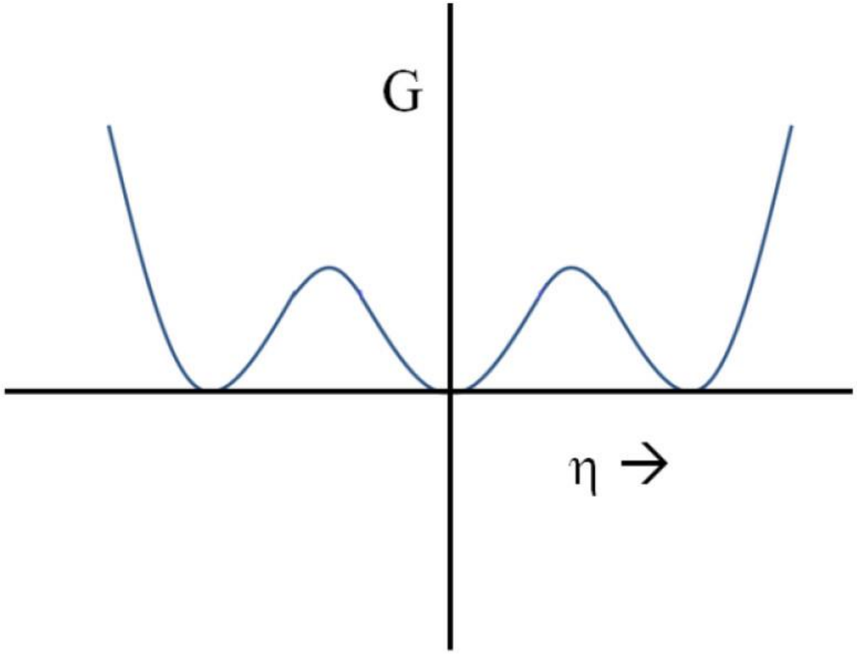
on eliminating $a(T\text{-}T_C)$ we get $\eta = 0 \text{ and } \eta^2 = -\dfrac{C}{2E}$

Solution to 15.2

Thus $$-\frac{2a(T-T_C)}{C}=-\frac{C}{2E}\ \text{(note for part d and } 4AE=C^2\text{)}$$

$$T=T_C+\frac{C^2}{4aE}$$

(b)



Solution to 15.2

(c)

Since there are two phases at $T = T_C + \dfrac{C^2}{4aE}$ with the same Gibbs energy and both are minima, this is a first order transformation. This value of T can be called T_0 in line with the way we have defined T_0 in the text.

(d)

$$\Delta H = T_{\mathrm{tr}} S^{XS}$$

$$\Delta S = S_{\mathrm{ordered}} - S_{\mathrm{disordered}}$$

$$\left(\frac{\partial G^{XS}}{\partial T}\right)_{T_{\mathrm{tr}}} = -\left(S^{XS}\right)_{T_{\mathrm{tr}}}$$

Solution to 15.2

$$G^{XS} = a\left(T - T_C\right)\eta_0^2 + C\eta_0^4 + E\eta_0^6$$

$$\left(\frac{\partial G^{XS}}{\partial T}\right)_{T_{\mathrm{tr}}} = a\eta_0^2 = -\left(S^{XS}\right)_{T_{\mathrm{tr}}}$$

$$\Delta H = -T_{\mathrm{tr}} a\eta_0^2 < 0$$

$$\Delta H = T_{\mathrm{tr}} a\left(\frac{C}{2E}\right) < 0 \quad (\text{since } C < 0)$$

(e)

It is an Exothermic transformation

15.3* 2-4 Landau Case

Using the equation

$$G^{\mathrm{XS}}(\eta) = a\left(T - T_0\right)\eta^2 + C\eta^4$$

show that the excess enthalpy for the Landau model with *B* = 0 and *C* > 0 is

$$\Delta H^{\mathrm{XS}} = \frac{a}{2}\frac{\left(T^2 - T_C^2\right)}{T_C}$$

Solution to 15.3

From Eq.5.10 that $$\eta_{\mathrm{eq}}^2 = \frac{T_C - T}{T_C} = 1 - \frac{T}{T_C}$$

Inserting $\eta_{\mathrm{eq}}^2 = \frac{T_C - T}{T_C}$ and $T_C = \frac{2C}{a}$ into $G^{\mathrm{XS}}(\eta) = a\left(T - T_0\right)\eta^2 + C\eta^4$

$$G^{\mathrm{XS}} = -\frac{a\left(T_C - T\right)^2}{2T_C} \quad \text{and} \quad -S^{\mathrm{XS}} = \frac{\partial G^{\mathrm{XS}}}{\partial T} = \frac{a\left(T_C - T\right)}{T_C}$$

$$H^{\mathrm{XS}} = G^{\mathrm{XS}} + TS^{\mathrm{XS}} = -\frac{a\left(T_C - T\right)^2}{2T_C} - \frac{Ta\left(T_C - T\right)}{T_C}$$

$$H^{\mathrm{XS}} = -\frac{a\left(T_C^2 - 2TT_C + T^2\right)}{2T_C} - \frac{2aTT_C - 2aT^2}{2T_C} \qquad H^{\mathrm{XS}} = \frac{a\left(T^2 - T_C^2\right)}{2T_C}$$

It can be seen that $\frac{dH^{\mathrm{XS}}}{dT} = C_P^{XS} = \frac{aT}{T_C}$ at T_C, thus $C_P^{XS} = a$

15.4* A solid is held at high temperature until equilibrium is attained. Its surface displays grooves, as shown in Figure 15.19.

a. Write an expression for the relationships between the grain boundary energy of α_1 and α_2.
b. Which grain boundary has the largest energy, α_1/α_2 or α_2/α_3?
c. If ϕ_{ij} goes to π, what is the value of the grain boundary energy?
d. If ϕ_{ij} goes to 0, what is the value of the grain boundary energy?

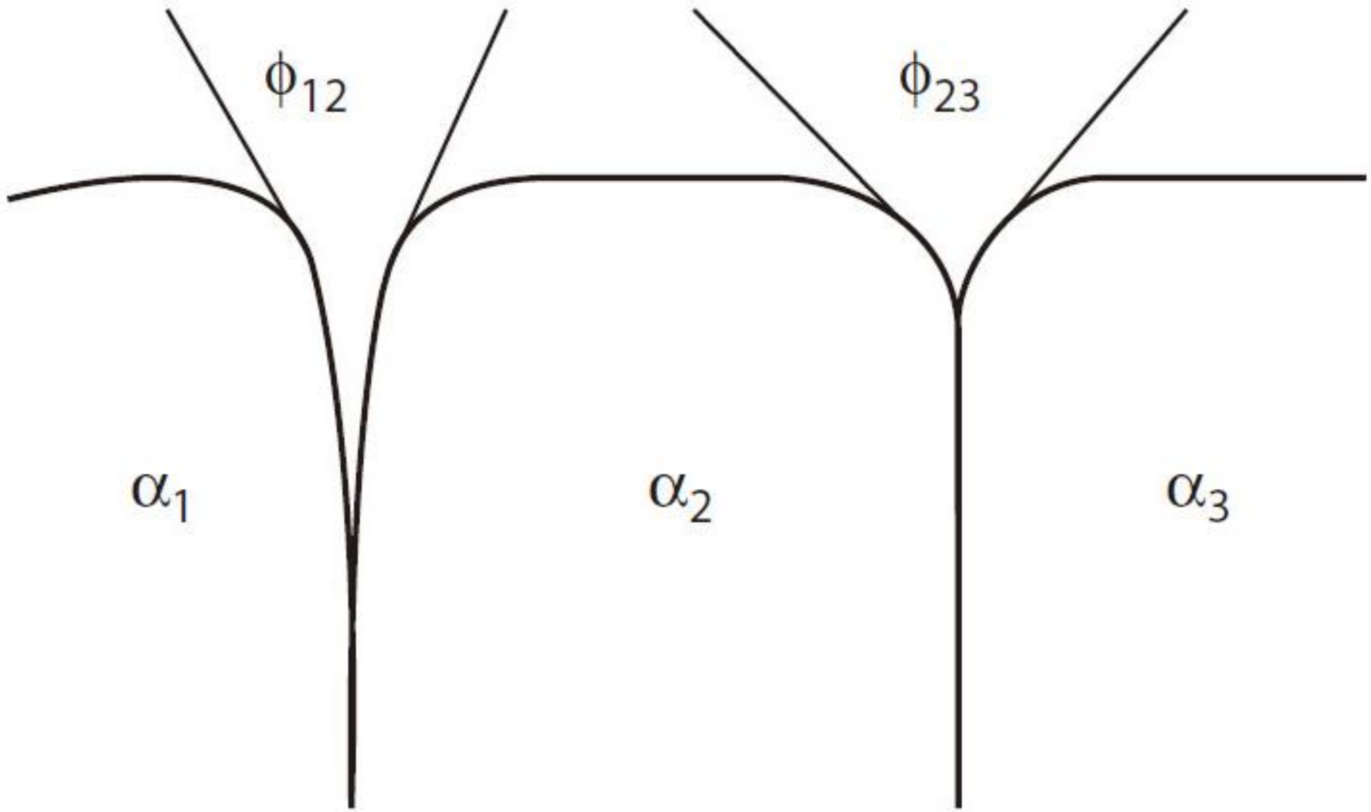


Figure 15.19 Grain boundary grooves arising from grain boundaries with different grain boundary energies.

Solution to 15.4

(a)

See problem 2.12* $\gamma_{gb/1-2} = 2\gamma_{\alpha_1/\alpha_2} \cos\left(\frac{\phi_{12}}{2}\right)$

(b)

The one with the smallest groove angle. Hence α_1/α_2

$$\frac{\gamma_{gb/1-2}}{\gamma_{gb/2-3}} = \frac{2\gamma_{\alpha_1/\alpha_2} \cos\left(\frac{\phi_{12}}{2}\right)}{2\gamma_{\alpha_2/\alpha_3} \cos\left(\frac{\phi_{23}}{2}\right)}$$

(c) $\gamma_{gb/i-j} = 2\gamma_{\alpha_i/\alpha_j} \cos\left(\frac{\phi_{ij}}{2}\right) \rightarrow 0$

(d) $\gamma_{gb/i-j} = 2\gamma_{\alpha_i/\alpha_j} \cos\left(\frac{\phi_{ij}}{2}\right) \rightarrow 2\gamma_{\alpha_i/\alpha_j}$

15.5* Small cylindrical particles have been observed to nucleate in certain alloy systems.

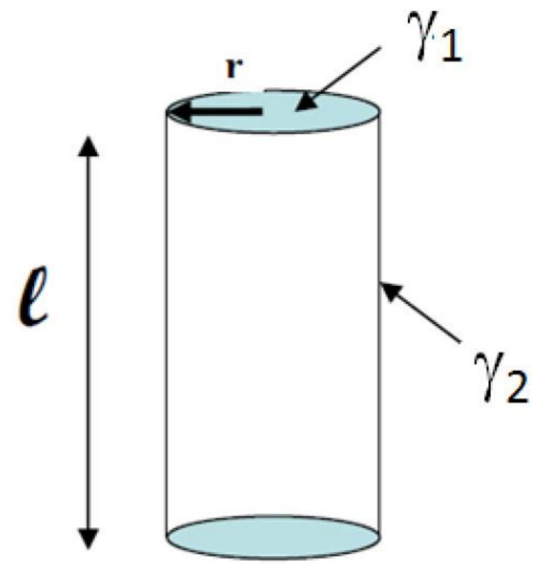
a. What values of *r* and *l* will minimize the energy barrier to the formation of these particles?

b. What surface energies favor the formation of long, thin cylinders? Explain.

- γ_1 is the surface energy of the circular face.
- γ_2 is the surface energy along the length of the cylinder.

Note: Assume the volume of the particle is constant.

Solution to 15.5



$$V = \pi r^2 \cdot l = \text{constant}$$

$$S.E. = \text{Surface Energy} = 2\pi r^2 \gamma_1 + 2\pi r l \gamma_2$$

$$S.E. = 2\pi r^2 \gamma_1 + \frac{V 2\pi r \gamma_2}{\pi r^2} = 2\pi r^2 \gamma_1 + \frac{V 2\gamma_2}{\pi r}$$

$$\frac{d\left(S.E.\right)}{dr} = 0 = 4\pi r \gamma_1 - V\frac{2\gamma_2}{\pi r^2}$$

$$\frac{r^*}{l^*} = \frac{1}{\pi}\frac{\gamma_2}{\gamma_1} \text{ for small } \gamma_1 \text{ large } r$$

Long rod for small γ_2 or large γ_1

Thin disc for large γ_2 or small γ_1

15.6* 2-6 Landau Case

The Gibbs energy of a phase can be written in terms of its order parameter as

$$G = a\left(T - T_C\right)\eta^2 + E\eta^6$$

where $a > 0$ and $E > 0$

a. What is the temperature T_0 where the disordered phase has the same Gibbs energy as the equilibrium ordered phase? Show work.

b. What is the value of the order parameter at T_0?

c. Is this a first-order or higher-order phase transition? Explain.

d. Show mathematically that, for $T < T_C$, the disordered phase is unstable.

Solution to 15.6

(a)

$G = 0$ for $\eta = 0$ (disordered)

And for the ordered phase $G = 0 = a\left(T - T_C\right)\eta^2 + E\eta^6$

or $$\eta^4 = -\frac{a\left(T - T_C\right)}{E}$$

Also since in equilibrium: $$\frac{\partial G}{\partial \eta} = 0 = 2a\left(T - T_C\right)\eta + 6E\eta^5$$

or $$\eta^4 = -\frac{a\left(T - T_C\right)}{3E}$$

For both to be true: $$T_0 = T_C$$

Solution to 15.6

(b)

The value is 0: that is, the phase is just beginning to order.

(c)

Must be a higher order transition

(d)

$$G = a\left(T - T_C\right)\eta^2 + E\eta^6$$

$$\frac{\partial G}{\partial \eta} = 2a\left(T - T_C\right)\eta + 6E\eta^5$$

$$\frac{\partial^2 G}{\partial \eta^2} = 2a\left(T - T_C\right) + 30E\eta^4 = 2a\left(T - T_C\right) \text{ for } \eta = 0$$

Thus $\dfrac{\partial^2 G}{\partial \eta^2} = 2a\left(T - T_C\right) < 0 \text{ for } T < T_C$

Therefore unstable (a maximum).

The End